# Nonlinear control of a 2 degree-of-freedom gimbal system

Haydar Kerem Karhan, Erkan Zergeroğlu

***Abstract*—This paper presents the nonlinear control of a 2 degree-of-freedom (DoF) gimbal system, where azimuth and elevation angles are controlled via interconnected joints. The system dynamics are derived using the Modified Denavit-Hartenberg method and expressed in a linearly parameterizable form. Three control strategies—robust, adaptive, and sliding mode control—are designed to ensure stability under parameter uncertainties and external disturbances. Stability analyses are performed using Lyapunov functions, demonstrating uniform asymptotic or exponential stability for each controller. Numerical simulations validate the effectiveness of the proposed strategies, showing accurate trajectory tracking and robustness against uncertainties. Future work includes hardware implementation and task space trajectory tracking.**

***Index Terms*—Nonlinear control systems, Robust control, Adaptive control**

## I. Introduction

GIMBAL systems are widely utilized in applications requiring precise orientation control, such as camera stabilization, robotics, and aerospace systems. Their inherent nonlinearities, coupled with parameter uncertainties and external disturbances, make them an ideal testbed for evaluating advanced control strategies. This paper focuses on the nonlinear control of a 2 degree-of-freedom (DoF) gimbal system, where the azimuth and elevation angles are controlled by two interconnected joints. The system dynamics are derived using the Modified Denavit-Hartenberg method, ensuring an accurate representation of its kinematics and dynamics.

The control objective is to ensure stable and robust performance under varying conditions, including parameter uncertainties. To achieve this, a series of control strategies are proposed and analyzed. These strategies include robust control, adaptive control, and sliding mode control, each tailored to handle specific challenges associated with the nonlinear dynamics of the gimbal system.

The remainder of this paper is organized as follows.
Section 2 provides a detailed description of the gimbal system's dynamics, including its linearly parameterizable representation and associated parameters. The equations of motion are derived, and the physical parameters are tabulated. In Section 3, the control design methodology is presented, starting with the exact model knowledge case, followed by robust control formulations and adaptive control schemes. The stability of each controller is analyzed using Lyapunov-based methods. Section 4 contains numerical simulations to evaluate the effectiveness of the proposed control strategies. The results are analyzed to demonstrate the performance of the controllers under different operating conditions. Section 5 concludes the findings and discusses potential directions of the study for future research.

## II. System Model

### A. Dynamics

The system used in this report is a 2 degree-of-freedom (dof) gimbal system where the azimuth and elevation are controlled by the consecutive joints. All kinematics and dynamics can be obtained through the script titled "gimbal_dynamics.mlx". The system is described using the Modified Denavit-Hartenberg method, and the kinematic and dynamic equations are obtained using [1]. The dynamics are presented in (1).

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau \tag{1}$$

Where $q(t), \dot{q}(t), \ddot{q}(t) \in \mathbb{R}^2$ represent the joint position, velocity and accelerations respectively, $M(q) \in \mathbb{R}^{2\times2}$ represents the inertia matrix, $C(q,\dot{q}) \in \mathbb{R}^{2\times2}$ is the Coriolis and centripetal force matrix, $G(q) \in \mathbb{R}^2$ is the vector containing gravity effects and $\tau \in \mathbb{R}^2$ is the input torque. The inertia matrix, Coriolis and centripetal matrix and the gravity vector are given in (2), (3) and (4) respectively.

$$M(q) = \begin{bmatrix} \theta_1 \cos^2(q_2) + \theta_2 & \theta_3 \sin(q_2) \\ \theta_3 \sin(q_2) & \theta_4 \end{bmatrix} \tag{2}$$

$$C(q,\dot{q}) = \begin{bmatrix} -\frac{1}{2}\theta_1 \sin(2q_2)\,\dot{q}_2 & -\frac{1}{2}\theta_1 \sin(2q_2)\,\dot{q}_1 + \theta_3 \cos(q_2)\,\dot{q}_2 \\ \frac{1}{2}\theta_1 \sin(2q_2)\,\dot{q}_1 & 0 \end{bmatrix} \tag{3}$$

$$G(q) = \begin{bmatrix} 0 \\ \theta_5 \cos(q_2) \end{bmatrix} \tag{4}$$

Where $q_1(t), q_2(t), \dot{q}_1(t), \dot{q}_2(t) \in \mathbb{R}$ represent the elements of the joint position and velocity vectors. $\theta_i \in \mathbb{R}$ represent the combined parameters of the system, which are presented in (5).

$$\begin{aligned} \theta_1 &= m_2 h_{2x}^2 - I_{xx2} + I_{yy2} \\ \theta_2 &= m_1 h_{1y}^2 + m_2 h_{2z}^2 + I_{zz1} + I_{xx2} \\ \theta_3 &= -m_2 h_{2x} h_{2z} + I_{xz2} \\ \theta_4 &= m_2 h_{2x}^2 + I_{zz2} \\ \theta_5 &= -m_2 h_{2x} g \end{aligned} \tag{5}$$

The values of the parameters are presented in Table 1, using SI units. Writing the dynamics using the combined parameters makes the dynamics presented in (1) linearly parametrizable, as shown in (5) in terms of $\theta \in \mathbb{R}^5$ where $\theta$ is the vector containing the combined parameters of the system.

$$Y\theta = M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) \tag{6}$$

TABLE 1 System parameters

| Parameter | Value |
|---|---|
| $m_1$ | $0.051\ [kg]$ |
| $m_2$ | $0.0195\ [kg]$ |
| $I_{zz1}$ | $2.084E-5\ [kg\ m^2]$ |
| $I_{xx2}$ | $7.4E-7\ [kg\ m^2]$ |
| $I_{yy2}$ | $3.86E-6\ [kg\ m^2]$ |
| $I_{zz2}$ | $3.75E-6\ [kg\ m^2]$ |
| $I_{xz2}$ | $-5.6E-7\ [kg\ m^2]$ |
| $h_{1y}$ | $-0.0227\ [m]$ |
| $h_{2x}$ | $0.0246\ [m]$ |
| $h_{2z}$ | $0.0014\ [m]$ |
| $g$ | $-9.80665\ [m\ s^{-2}]$ |

Furthermore, a Simscape Multibody design in Simulink is implemented to have visual simulations of the designed controllers. This is shown in Figure

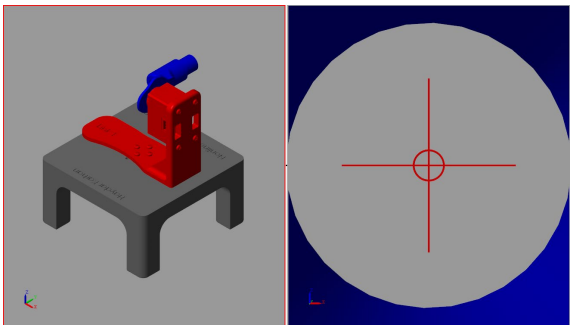


Fig. 1 Simscape Multibody design

A hardware setup was also developed. However, due to time constraints, the real system wasn't tested. In future works, the real system will be used to track task space trajectories.

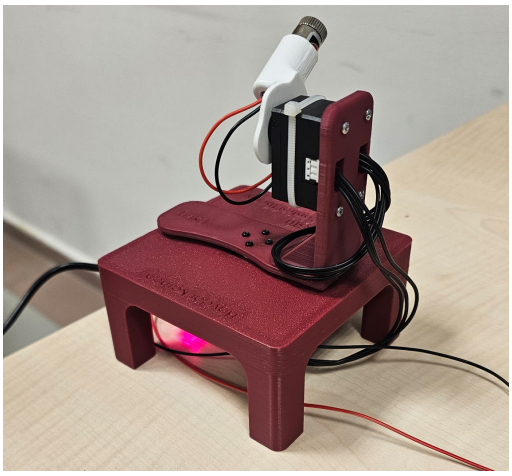


Fig. 2 Real system

All parameters' values are obtained through measuring the actual system parameters, and using Solidworks to obtain center of mass and inertia matrix values. Only the friction is not used in the model, since the hardware setup wasn't used to estimate friction values. Therefore, friction is neglected in the analysis.

## III. Controller Design

The controllers will be designed in the joint space of the system. Therefore, no task space trajectories will be directly tracked. If a task space trajectory is to be tracked, the task space trajectory must first be converted to the joint space.

### A. Exact Model Knowledge

Provided that all parameters in the model are known, the controller design is straightforward. Introducing the joint space tracking error $e(t) \in \mathbb{R}^2$ as

$$e = q_d - q \tag{7}$$

Define a filtered tracking error to incorporate the first and second derivatives in a single term as

$$r = \dot{e} + \alpha e \tag{8}$$

Where $\alpha \in \mathbb{R}^{2\times2}$ is a positive definite, diagonal matrix. Note that as $r(t) \to 0 \Longrightarrow e(t) \to 0,\ \dot{e}(t) \to 0$.

$$\begin{gathered} \dot{r} = \ddot{e} + \alpha\dot{e} \\ \dot{r} = \ddot{q}_d - \ddot{q} + \alpha\dot{e} \\ M\dot{r} = M\ddot{q}_d - M\ddot{q} + M\alpha\dot{e} \\ M\dot{r} = M(\ddot{q}_d + \alpha\dot{e}) + C\dot{q} + G - \tau \end{gathered} \tag{9}$$

Writing $\dot{q} = \dot{q}_d + \alpha e - r$ yields

$$\begin{gathered} M\dot{r} = M(\ddot{q}_d + \alpha\dot{e}) + C(\dot{q}_d + \alpha e) + G - Cr - \tau \\ M\dot{r} = N - Cr - \tau \\ N = M(\ddot{q}_d + \alpha\dot{e}) + C(\dot{q}_d + \alpha e) + G \end{gathered} \tag{10}$$

The following Lyapunov function is introduced to design the controller.

$$\begin{gathered} V = \frac{1}{2} r^\top M r \\ \dot{V} = \frac{1}{2} r^\top \dot{M} r + r^\top M \dot{r} \\ \dot{V} = \frac{1}{2} r^\top \dot{M} r + r^\top (N - Cr - \tau) \\ \dot{V} = \frac{1}{2} r^\top \big(\dot{M} - 2C\big) r + r^\top (N - \tau) \end{gathered} \tag{11}$$

The inertia matrix and the Coriolis and centripetal force matrix satisfy the following skew symmetric property

$$\xi^\top \big(\dot{M} - 2C\big)\xi = 0 \tag{12}$$

Therefore, the time derivative of the Lyapunov function simplifies to

$$\dot{V} = r^\top (N - \tau) \tag{13}$$

Choosing the control input as

$$\tau = K_r r + M(\ddot{q}_d + \alpha\dot{e}) + C(\dot{q}_d + \alpha e) + G \tag{14}$$

yields the following function where $K_r \in \mathbb{R}^2$ is a positive definite diagonal matrix

$$\dot{V} = -r^\top K_r r \tag{15}$$

Which is a negative definite, function, showing that the chosen controller ensures uniform asymptotic stability

### B. Robust Control

When the system contains parameter uncertainties, exact model knowledge is not available. Therefore, the controller must be designed considering the uncertainties in the system. To do this, the same filtered tracking error will be used, which leads to the same dynamics as in (10).

Define $\widehat{N}$ as the estimated dynamics, and $\widetilde{N} = N - \widehat{N}$ is the error between the real and estimated dynamics. The robust controller formulation also assumes that $\widetilde{N} \le \rho(\|r\|)$.
Using the Lyapunov function

$$V = \frac{1}{2} r^\top M r \tag{16}$$

It's time derivative becomes

$$\dot{V} = r^\top (N - \tau) \tag{17}$$

Upon choosing the control as

$$\tau = K_r r + \widehat{N} + V_R \tag{18}$$

Where $V_R$ is a robust term chosen by the designer. The time derivative of the Lyapunov function now becomes

$$\dot{V} = r^\top \big(\widetilde{N} - K_r r - V_R\big) \tag{19}$$

Now, upon choosing the robust term $V_R$, different types of robust controller formulations are presented

#### 1) High frequency type

The high frequency type chooses the robust term as

$$V_R = \frac{\rho^2}{\rho\|r\| + \varepsilon} r \tag{20}$$

Where the function in (19) becomes

$$\dot{V} = -r^\top K_r r + r^\top \widetilde{N} - r^\top \frac{\rho^2}{\rho\|r\| + \varepsilon} r \tag{21}$$

Which can be upper bounded as:

$$\begin{aligned}\dot{V} &\leq -\lambda_{min}\{K_r\}\|r\|^2 + \rho\|r\| - \frac{\rho^2\|r\|^2}{\rho\|r\| + \varepsilon} \\ \dot{V} &\leq -\lambda_{min}\{K_r\}\|r\|^2 + \rho\|r\|\left(1 - \frac{\rho\|r\|}{\rho\|r\| + \varepsilon}\right) \\ \dot{V} &\leq -\lambda_{min}\{K_r\}\|r\|^2 + \varepsilon\left(\frac{\rho\|r\|}{\rho\|r\| + \varepsilon}\right)\end{aligned} \tag{22}$$

Where in the worst case, the term $\left(\frac{\rho\|r\|}{\rho\|r\|+\varepsilon}\right) = 1$. Thus, the function can be bound as

$$\dot{V} \leq -\lambda_{min}\{K_r\}\|r\|^2 + \varepsilon \tag{23}$$

Which shows that the designed controller is ultimately uniformly bounded.

#### 2) High gain type

For the high gain type, the robust term is chosen as

$$V_R = \frac{\rho^2}{\varepsilon} r \tag{24}$$

The time derivative of the function presented in (16) becomes

$$\dot{V} = -r^\top K_r r + r^\top \tilde{N} - r^\top \frac{\rho^2}{\varepsilon} r \tag{25}$$

Which can be bounded as

$$\begin{aligned}\dot{V} &\leq -\lambda_{min}\{K_r\}\|r\|^2 + \rho\|r\| - \frac{\rho^2\|r\|^2}{\varepsilon} \\ \dot{V} &\leq -\lambda_{min}\{K_r\}\|r\|^2 + \rho\|r\|\left(1 - \frac{\rho\|r\|}{\varepsilon}\right)\end{aligned} \tag{26}$$

Where now the function can be bounded in 2 ways,

$$\rho\|r\| > \varepsilon \rightarrow \dot{V} = -\lambda_{min}\{K_r\}\|r\|^2 \tag{27}$$

$$\varepsilon > \rho\|r\| \rightarrow \dot{V} = -\lambda_{min}\{K_r\}\|r\|^2 + \varepsilon \tag{28}$$

Where the case presented in (27) gives exponential stability, and the case presented in (28) gives ultimately uniformly boundedness.

#### 3) Sliding mode type

In this sliding mode type, the robust term is chosen as

$$V_R = \rho\, sgn(r) \tag{29}$$

Where $sgn$ is the signum function defined as

$$sgn(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases} \tag{30}$$

If the input is a vector, the function operates on each element of the vector.

$$\begin{aligned}\dot{V} &= -r^\top K_r r + r^\top \tilde{N} - r^\top \rho\, sgn(r) \\ \dot{V} &\leq -\lambda_{min}\{K_r\}\|r\|^2 + \rho\|r\| - \rho\, r^\top sgn(r)\end{aligned} \tag{31}$$

Since $r^\top sgn(r) \geq 0$ the function is bound as

$$\dot{V} \leq -\lambda_{min}\{K_r\}\|r\|^2 \tag{32}$$

Thus, the designed control is exponentially stable.

### C. Adaptive Control

#### 1) Standard adaptive

Linearly parameterizing $N$ in the general form given below can be done as follows

$$N = Mu + Cv + G = Y\theta \tag{33}$$

Where $u(t), v(t) \in \mathbb{R}^2$ and $Y \in \mathbb{R}^{2\times 5}$.

$$\begin{aligned} Y &= \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} & 0 & 0 \\ Y_{21} & 0 & Y_{23} & Y_{24} & Y_{25} \end{bmatrix} \\ Y_{11} &= u_1 \cos^2(q_2) - \frac{1}{2}\sin(2q_2)\,(v_1\dot{q}_2 + v_2\dot{q}_1) \\ Y_{21} &= \frac{1}{2} v_1 \sin(2q_2)\,\dot{q}_1 \\ Y_{12} &= u_1 \\ Y_{13} &= u_2 \sin(q_2) + v_2 \cos(q_2)\,\dot{q}_2 \\ Y_{23} &= u_1 \sin(q_2) \\ Y_{24} &= u_2 \\ Y_{25} &= \cos(q_2) \end{aligned} \tag{34}$$

We can write our estimated parameters as $\hat{\theta}$ and the error between the real parameters and the estimate as $\tilde{\theta} = \theta - \hat{\theta}$. Defining the following Lyapunov function

$$\begin{aligned} V &= \frac{1}{2} r^\top M r + \frac{1}{2}\tilde{\theta}^\top \Gamma^{-1}\tilde{\theta} \\ \dot{V} &= r^\top (N - \tau) + \tilde{\theta}^\top \Gamma^{-1}\left(\dot{\theta} - \dot{\hat{\theta}}\right) \end{aligned} \tag{35}$$

Since the true parameters are time invariant, $\dot{\theta} = 0$. Thus, the function becomes:

$$\dot{V} = r^\top (Y\theta - \tau) - \tilde{\theta}^\top \Gamma^{-1}\dot{\hat{\theta}} \tag{36}$$

Choosing the parameter adaptation law and the controller as follows, gives the following Lyapunov function time derivative

$$\begin{aligned} \dot{\hat{\theta}} &= \Gamma Y^\top r \\ \tau &= K_r r + Y\hat{\theta} \end{aligned} \tag{37}$$

$$\dot{V} = r^{\top}\left(Y\theta - K_r r - Y\hat{\theta}\right) - \tilde{\theta}^{\top}\Gamma^{-1}\Gamma Y^{\top} r$$

$$\dot{V} = r^{\top}\left(Y\theta - K_r r - Y\hat{\theta}\right) - r^{\top} Y\tilde{\theta}$$

$$\dot{V} = r^{\top}\left(Y\tilde{\theta} - K_r r\right) - r^{\top} Y\tilde{\theta} \tag{38}$$

$$\dot{V} = -r^{\top} K_r r$$

Thus, the designed control and adaptation law are asymptotically stable.

*2) Desired model compensation*

Writing the desired dynamics is done by

$$N_d = M\ddot{q}_d + C\dot{q}_d + G(\dot{q}_d) = Y_d\theta \tag{39}$$

Now, $M\dot{r}$ can be rewritten as

$$M\dot{r} = N - Cr - \tau - N_d + N_d$$

$$M\dot{r} = \mathcal{X} - Cr - \tau + N_d \tag{40}$$

Provided that $\mathcal{X}$ can be bounded as $\mathcal{X} \leq \rho(\|r\|)\|r\|$

Defining the Lyapunov function similarly to (35). The time derivative becomes

$$\dot{V} = r^{\top}(\mathcal{X} - \tau + N_d) - \tilde{\theta}^{\top}\Gamma^{-1}\dot{\hat{\theta}} \tag{41}$$

Introducing the control input and adaptation law as

$$\tau = K_r r + Y_d\hat{\theta} + \rho r$$

$$\dot{\hat{\theta}} = \Gamma Y_d^{\top} r \tag{42}$$

The time derivative of the Lyapunov function becomes

$$\dot{V} = r^{\top}\left(\mathcal{X} - K_r r - \rho r - Y_d\hat{\theta} + Y_d\theta\right) - \tilde{\theta}^{\top}\Gamma^{-1}\dot{\hat{\theta}}$$

$$\dot{V} = r^{\top}\left(\mathcal{X} - K_r r - \rho r + Y_d\tilde{\theta}\right) - r^{\top} Y_d\tilde{\theta}$$

$$\dot{V} = r^{\top}(-K_r r - \rho r + \mathcal{X}) \tag{43}$$

$$\dot{V} \leq -\lambda_{min}\{K_r\}\|r\|^2 - \rho\|r\|^2 + \rho\|r\|^2$$

$$\dot{V} \leq -\lambda_{min}\{K_r\}\|r\|^2$$

Therefore, the designed controller and update law make the system exponentially stable.

## IV. Simulation Results

The following results are for a sinusoidal input at a period of 2 seconds and an amplitude of 0.4 rad for each joint.

### A. Exact Model Knowledge

In this formulation, $\alpha$ was chosen to be $diag([1,1])$ and $K_r$ was chosen to be $diag([2,2])$.
The exact model knowledge case provides asymptotic stability shown in Fig. 3.

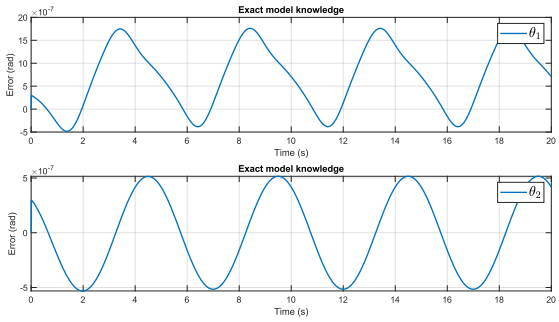



Fig. 3 Exact model knowledge error response

The response of the derivative of the error is shown in Fig. 4.

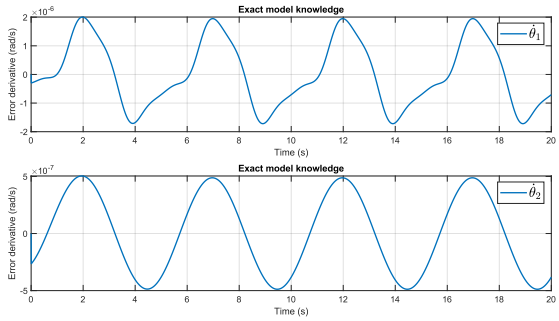



Fig. 4 Exact model knowledge error derivative response

Now, we introduce uncertainties in the model by adding a fixed mass of 0.014 kg to all masses in the system, 5e-6 kgm$^2$ of inertia to all inertias in the system and 0.0075 m of length to all lengths in the system. With these uncertainties in the system, exact model knowledge is not known, therefore the same controller performs worse as presented in Fig. 5

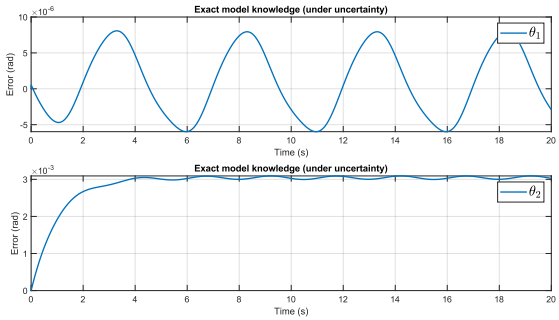



Fig. 5 Exact model knowledge based controller under uncertainties

The results of the exact model knowledge controllers with and without uncertainties are presented in Table 2.

TABLE 2 Exact model knowledge errors

| | $e_1\ [rms]$ | $e_2\ [rms]$ | $\dot{e}_1\ [rms]$ | $\dot{e}_2\ [rms]$ |
|---|---|---|---|---|
| E.m.k | 9.91e-7 | 3.65e-7 | 1.14e-6 | 3.46e-7 |
| E.m.k (uncertainty) | 4.83e-6 | 2.93e-3 | 6.56e-6 | 4.84e-4 |

## B. Robust Controller

### 1) High gain type

For the high gain type controller, $\alpha$ was taken $diag([1,1])$, $K_r$ is $diag([10,10])$, $\rho$ is 1 and $\varepsilon$ is 0.01.
The results for the high gain type robust controller designed are presented in Fig. 6 and Fig. 7.

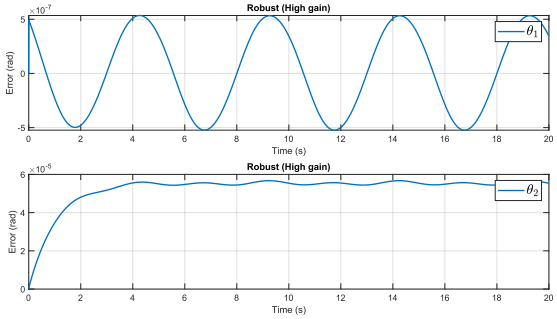



Fig. 6 High gain type robust control error

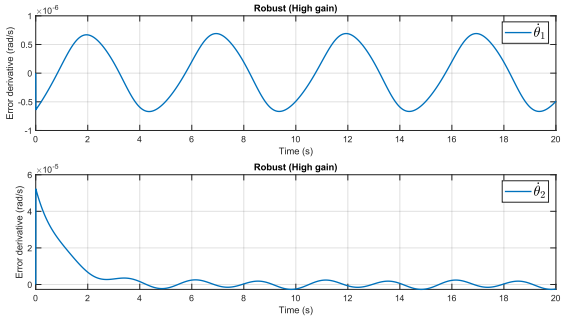



Fig. 7 High gain type robust control error derivative

### 2) High frequency type

For the high frequency type controller, $\alpha$ was taken $diag([1,1])$, $K_r$ is $diag([10,10])$, $\rho$ is 1 and $\varepsilon$ is 0.01.

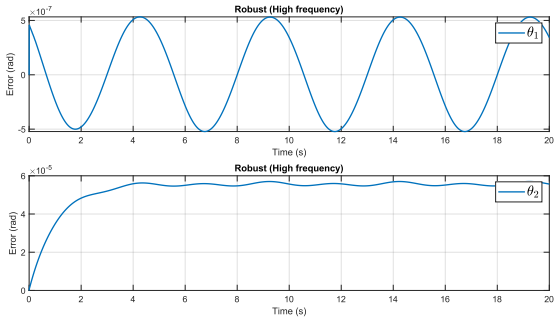



Fig. 8 High frequency type robust control error

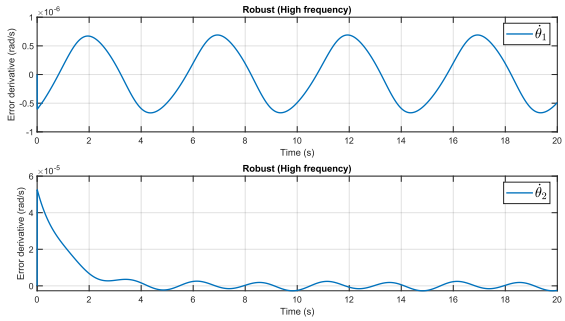



Fig. 9 High gain type robust control error derivative

### 3) Sliding mode

For the sliding mode controller, $\alpha$ was taken $diag([1,1])$, $K_r$ is $diag([10,10])$, $\rho$ is $diag([0.1\ 0.1])$.

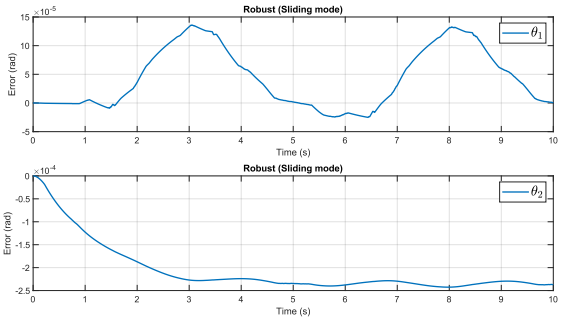



Fig. 10 Sliding mode robust control error

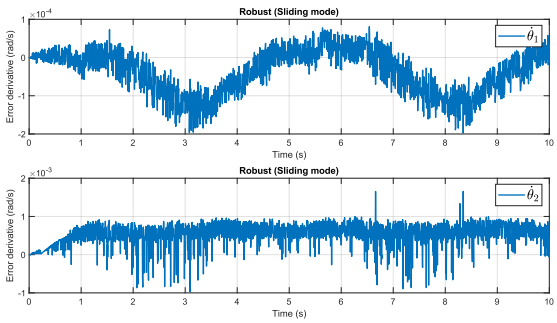



Fig. 11 Sliding mode robust control error derivative

The results of the robust control designs are presented in Table 3.

TABLE 3 Robust control errors

| | $e_1\ [rms]$ | $e_2\ [rms]$ | $\dot{e}_1\ [rms]$ | $\dot{e}_2\ [rms]$ |
|---|---|---|---|---|
| High gain | 3.75e-7 | 5.33e-5 | 4.77e-7 | 8.80e-6 |
| High frequency | 3.74e-7 | 5.35e-5 | 4.76e-7 | 8.85e-6 |
| Sliding mode | 6.72e-5 | 2.14e-4 | 6.78e-5 | 6.08e-4 |

## C. Adaptive Control

### 1) Standard adaptive

For the standard adaptive control, $\alpha$ was taken $diag([1,1])$, $K_r$ is $diag([10,10])$ and $\Gamma$ is $diag([5,5,5,5,5])$. Furthermore, the numerical integrator used to compute $\hat{\theta}$ from $\dot{\hat{\theta}}$ is initiliazed with the nominal parameters $\theta_{1\ldots5}$. The results are presented in Fig. 12.

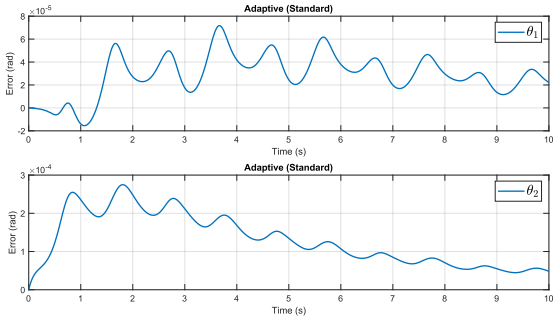



Fig. 12 Standard adaptive control error

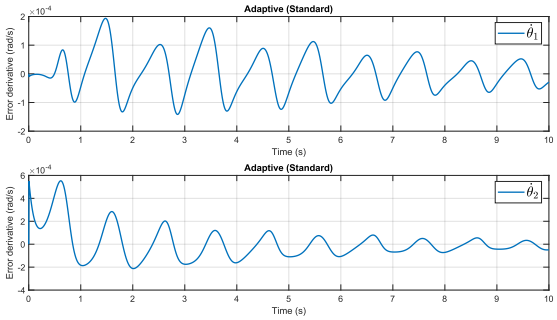



Fig. 13 Standard adaptive control error derivative

### 2) Desired model compensation

For the desired model compensation adaptive control, $\alpha$ was taken $diag([1,1])$, $K_r$ is $diag([10,10])$ and $\Gamma$ is $diag([5,5,5,5,5])$. Furthermore, $V_R$ was taken as $\rho\|r\|$ with $\rho$ being 0.1. Additionally, the numerical integrator used to compute $\hat{\theta}$ from $\dot{\hat{\theta}}$ is initiliazed with the nominal parameters $\theta_{1\ldots5}$. The results are presented in Fig. 14.

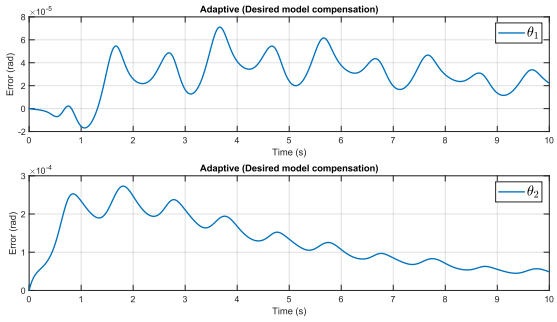



Fig. 14 Desired model compensation error

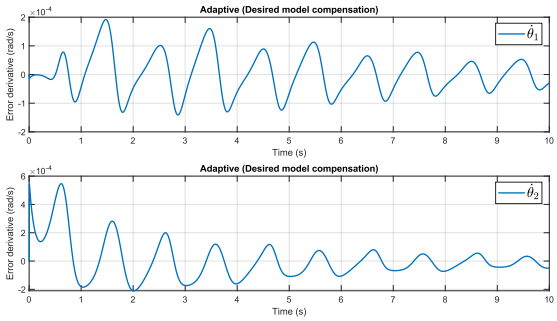



Fig. 15 Desired model compensation error derivative

The results from both adaptive control structures are presented in Table 4.

TABLE 4 Adaptive control errors

| | $e_1\ [rms]$ | $e_2\ [rms]$ | $\dot{e}_1\ [rms]$ | $\dot{e}_2\ [rms]$ |
|---|---|---|---|---|
| Adaptive | 3.43e-5 | 1.47e-4 | 7.03e-5 | 1.37e-4 |
| Desired model comp | 3.40e-5 | 1.46e-4 | 7.02e-5 | 1.36e-4 |

## V. Conclusions and Further Works

The designed controllers all performed well in terms of their respective rms error values. With uncertainties taken into account however, the robust high gain and high frequency performed the best. These results are presented in Table 5.

In the simulations, actuator dynamics and friction were neglected. Therefore, for the analysis to work on the designed real system, viscous friction and Coulomb friction models must be used to represent the real system correctly. Furthermore, actuator dynamics must also be present in the model in order to account for acceleration/deceleration limits, velocity constraints and actuator torque.

However, in the case the analysis is performed assuming that the required torque is directly supplied, the controllers all perform reasonably.

TABLE 5 Controller comparisons

| | $e_1\ [rms]$ | $e_2\ [rms]$ | $\dot{e}_1\ [rms]$ | $\dot{e}_2\ [rms]$ |
|---|---|---|---|---|
| E.m.k | 9.91e-7 | 3.65e-7 | 1.14e-6 | 3.46e-7 |
| E.m.k (uncertainty) | 4.83e-6 | 2.93e-3 | 6.56e-6 | 4.84e-4 |
| High gain | 3.75e-7 | 5.33e-5 | 4.77e-7 | 8.80e-6 |
| High frequency | 3.74e-7 | 5.35e-5 | 4.76e-7 | 8.85e-6 |
| Sliding mode | 6.72e-5 | 2.14e-4 | 6.78e-5 | 6.08e-4 |
| Adaptive | 3.43e-5 | 1.47e-4 | 7.03e-5 | 1.37e-4 |
| Desired model comp | 3.40e-5 | 1.46e-4 | 7.02e-5 | 1.36e-4 |

## VI. Bonus

Task space tracking was also performed by converting the known task space trajectory to a joint space trajectory using the following properties

$$q_2 = \text{atan2}\left(d_3, \sqrt{d_1^2 + d_2^2}\right)$$

$$q_1 = \text{atan2}\left(-\frac{d_1}{\cos(q_2)}, \quad \frac{d_2}{\cos(q_2)}\right)$$

Where $d = [d_1\ d_2\ d_3]^{\mathsf{T}}$ is the unit vector pointing from the end effector of the robot to the target in the task space, defined as:

$$d = r - P_{ee}$$

Where $r$ is the position of the target in the task space.
Generating the reference velocity and acceleration is done using the following equations

$$\dot{x} = J\dot{q}$$
$$\ddot{x} = \dot{J}\dot{q} + J\ddot{q}$$
$$\dot{q} = J^{-1}\dot{x}$$
$$\ddot{q} = J^{-1}\left(\ddot{x} - \dot{J}\dot{q}\right)$$

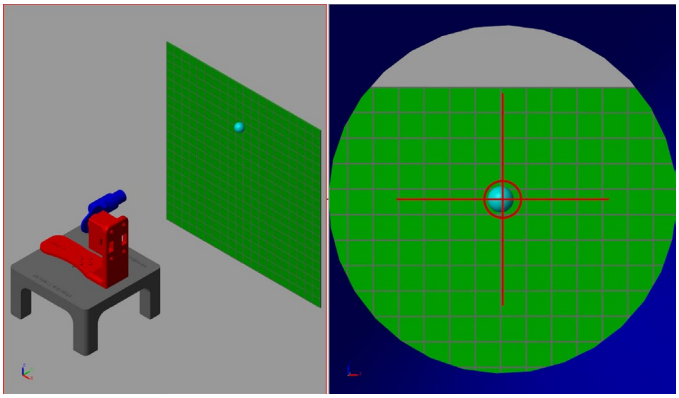


Fig. 16 Simscape task space target tracker